PRANAV KUMAR

B.Tech CSE • Full-Stack & AI/ML Engineer

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GitHub | LinkedIn | LeetCode

PROFESSIONAL SUMMARY

Computer Science undergraduate with a strong foundation in full-stack development and applied AI/ML, backed by two peer-reviewed research publications on transformer-based vulnerability prediction and deep-reinforcement-learning intrusion detection. Experienced in building production-style full-stack applications (React, Spring Boot, PostgreSQL) and training/evaluating deep learning models (PyTorch, HuggingFace). Currently expanding cloud and DevOps expertise across AWS, Google Cloud, and Kubernetes. Seeking internship opportunities in AI/ML, full-stack engineering, or technical consulting.

EDUCATION DETAILS

Course	Institute Name	University / Board	Year of Passing	Score (CGPA/%)	Course Type
B.Tech (CSE)	Kalinga Institute of Industrial Technology	KIIT (Deemed to be University)	Expected 2027	8.64 CGPA	Full-time
12th	FIITJEE Hyderabad	Telangana Board of Intermediate Education	2023	94.6%	Full-time
10th	Glendale Academy International School	CBSE	2021	96.4%	Full-time

PUBLICATIONS & RESEARCH

Rewiring Transformers for Exploit Likelihood [Paper] Dec 2024 – Apr 2025

Published in AIP Conference Proceedings (ADSSS) • HuggingFace, PyTorch

- Fine-tuned DistilBERT and GPT2-Large on 56,000+ NVD vulnerability records (2014–2024) to predict exploit likelihood from textual descriptions and numerical CVSS scores.
- DistilBERT achieved 85.6% accuracy, 92.1% precision, and 90.0% F1-score — outperforming GPT2-Large by 16.5%.
- Fully open-sourced the codebase and dataset pipeline on GitHub for reproducibility.
- Announced and shared findings publicly on LinkedIn, drawing engagement from the ML research community.

AI-Based Intrusion Detection for Wireless Sensor Networks Oct 2025 – Jan 2026

Published at ICNCDA 2026, University of Essex • PyTorch, Deep Reinforcement Learning • [Certificate]

- Designed IDFSWSN-DRL, combining Temporal Convolutional Networks with Multi-Head Attention and Gazelle Optimization Algorithm (GOA)-based feature selection, reducing the feature space by ~68%.
- Achieved 99.09% accuracy with strong precision and recall on the WSN-DS dataset, outperforming Naïve Bayes and Random Forest baselines.
- Enabled real-time detection of Blackhole, Grayhole, and Flooding attacks with low computational overhead, suitable for resource-constrained sensor deployments.

PROJECTS UNDERTAKEN

Diet Tracker — Full-Stack Nutrition Application

React, Spring Boot, PostgreSQL / SQLite, JWT Authentication, Docker-ready

- Built a full-stack nutrition tracking application with a React frontend (using useEffect hooks for live API integration) and a Spring Boot REST backend.

- Implemented macronutrient and calorie tracking, meal logging, JWT-based user authentication, and persistence via PostgreSQL/SQLite.
- Designed a fully responsive UI across mobile and desktop viewports, containerized for Docker-based deployment.
- Developed a companion frontend-only prototype featuring dark mode, SVG ring charts, and Recharts-based data visualizations.

STRONG SUBJECTS

Data Structures & Algorithms • Full-Stack Development • Machine Learning • Natural Language Processing • Artificial Intelligence • Cloud Computing • Cybersecurity • GenAI (LangChain, LangGraph)

CERTIFICATIONS & ACHIEVEMENTS

- AWS Cloud Computing Course — Udemy [Certificate]
- Full Stack Web Development — Udemy [Certificate]
- Data Structures & Algorithms (C/C++) — Udemy [Certificate]
- International Rank 1 — International English Olympiad, 2020–21 [Certificate]
- Google Cloud DevOps [Certificate]
- IBM GenAI and Prompt Engineering — Coursera [Certificate]

TECHNICAL SKILLS

Languages: C/C++, Java, JavaScript, Python, SQL, HTML5/CSS3

Frameworks & Libraries: React, Node.js, Express.js, Spring Boot, PyTorch, PostgreSQL, EJS

Cloud & DevOps: AWS, Google Cloud Platform (GCP), Google Cloud DevOps, Kubernetes (GKE), Docker, Git/GitHub, Linux/Ubuntu, VS Code, Cisco Packet Tracer

Areas of Expertise: Data Structures & Algorithms, Full-Stack Development, Machine Learning, NLP, Artificial Intelligence, Cloud Computing, GenAI (LangChain, LangGraph)

In Progress: ServiceNow CSA certification

ACTIVITIES & INTERESTS

- Played in the Hyderabad Cricket Association (HCA) One-Day League (A3) for Sports Zone Cricket Academy.
- Participated in EY Techathon 5.0 and Tutedude hackathons — managed API integrations and backend development.
- Football enthusiast with a deep interest in tactical analysis.